

HRRP Clutter Rejection Via One-Class Classifier With Hausdorff Distance

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Lots of detectors for high resolution range profile (HRRP) data are mainly based on the intensities of target echoes. When the intensities of outliers are as large as those of targets of interest, some false alarms will emerge during the detection stage. In this article, we propose a rejection algorithm for HRRP data between the detection stage and recognition stage to eliminate the false alarms occurring during the detection stage. In this method, the intensities and positions of the dominant scatterers are extracted as a joint feature. Then, the feature is used to develop the K-center one-class classifier based on Hausdorff distance. Experimental results on the measured HRRP data indicate that the proposed rejection algorithm can eliminate the false alarms remarkably well, and keep most of the target data as well. Meanwhile, the proposed method has better performance under different values of signal-to-noise ratio and different values of parameters than some other rejection methods.

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I. INTRODUCTION

Since HRRPs contain abundant structure details of targets [3], [4], they have been widely used in target detection and recognition [4]–[7], [12], [13], [16], [17], [23]. A target in an HRRP can be considered a range-spread target with its multiple scatterers spreading in different range cells. Range-spread target detection in high-resolution radars can go back to the early works of Hughes [23]. In [23], several detectors for range-spread targets are proposed based on energy integration. In [5], a heuristic detector is proposed to detect range-spread targets in white Gaussian noise using consecutive HRRPs. The detector contains a nonlinear shrinkage mapping process and a cross-correlation integrator. In [6], a method is proposed by using the low-rank prior knowledge of clutter covariance matrix to estimate parameters. In [12], three generalized likelihood ratio test detectors for range-spread target in grouped partially homogeneous backgrounds are developed. To boost radar performance, some advanced probing strategies, such as those in [24] and [25], are also conceived.

Many detectors more or less utilize the energy information and are proved optimal only in some specific kinds of clutter. Therefore, in some scenes with strong clutter, many false alarms may emerge in the outputs of these energy-based detectors. If the outputs containing many false alarms are directly used in recognition stage, the recognition algorithm will be designed to be complicated to maintain good recognition performance. Besides, the presence of false alarms leads to large amounts of computation. Therefore, a rejection stage should be adopted between detection and recognition stages.

The main goal of a rejection problem is to remove the outlier samples and meanwhile reserve the samples from targets of interest. Some rejection methods have been proposed [18]–[21]. In [18], a new open set recognition method is proposed, in which the outputs of a support vector machine are mapped to probabilities. In [19], a method for automatic target recognition is proposed, in which unclassifiable data is rejected to improve reliability. However, few rejection methods are specially proposed for HRRP data. A rejection process consists of feature extraction and classifier design.

Finding the distinct features is key to HRRP rejection. There are many distinctive features used for synthetic aperture radar (SAR) automatic target recognition. In [1] and [2], some methods based on local bright spots with abundant information are proposed for SAR target classification. An HRRP can be seen as the amplitude of the coherent summations of the complex time returns from target scatterers in each range cell [7]. For different kinds of targets or clutter, the scatterers distributions are always different, and so are the dominant scatterers distributions. Although HRRPs only contain the information in range domain, the dominant scatterers which are similar to the bright spots in SAR images also provide abundant information about structure details. For different classes of samples, the dominant scatterers commonly have different intensities and different distributions in positions. Therefore, the feature

based on intensities and positions of dominant scatterers are proposed for the HRRP rejection in this article. The HRRP recovery methods based on the sparse recovery [24], [25] can be used in the extraction of dominant scatterers. However, those methods commonly contain iterative process which brings large amount of calculation. In this article, the cell-average constant false alarm rate (CA-CFAR) [10], [11] detector is adopted to extract the dominant scatterers, which is easy to implement with small amounts of computation.

The selection of classifier is also key to a rejection process. In one-class classifiers, it is assumed that only target class is available in the training step. Since there are always so many kinds of natural and artificial clutter in complicated scenes, we cannot get enough outlier samples in the training step. Therefore, the one-class classifier is adopted. Traditional one-class classifiers are based on Euclidean distance, which cannot measure the distance between point sets, such as our proposed feature. The Hausdorff distance (HD) measure is a well-known method for representing the distance between two point sets [1], [14], [15]. In this article, we develop the K-center one-class classifier based on HD for our proposed feature.

The target-aspect sensitivity is a common issue in target rejection and recognition. In a high-resolution radar system, scatterers of a target are separated in several range cells. The scatterers distribution will vary with the change of target-aspect. In [4], [7], [13], [16], and [17], a relatively complete target dataset, which contains target echoes from almost all aspects, is used to train the classifier. In this article, we also use a relatively complete training dataset to train the one-class classifiers [9].

The contributions of this article are summarized as follows.

- 1) A feature containing the positions and intensities of dominant scatterers, which provide abundant information about structure details, are proposed for HRRP rejection.
- 2) The K-center one-class classifier is adopted in this article. The K-center based on HD is developed for our proposed feature.
- 3) The extraction of dominant scatterers suppresses the noise to some extent. The alignment method based on the dominant scatterers performs well against the noise.

The rest of this article is organized as follows. In Section II, feature extraction method based on CA-CFAR is introduced. In Section III, the HD and our rejection method are introduced. In Section IV, some experimental results based on measured data are given. Section V concludes this article.

II. FEATURE EXTRACTION BASED ON CA-CFAR

A. CA-CFAR Detector

Here, the CA-CFAR detector is adopted to extract the dominant scatterers. Fig. 1 shows the flowchart of CA-CFAR.

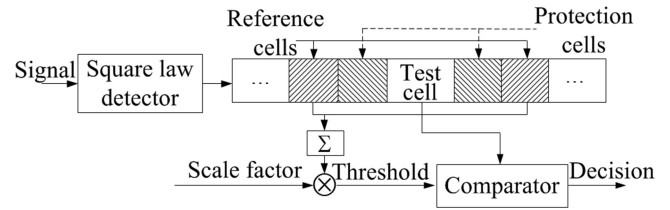


Fig. 1. Flowchart of CA-CFAR detector.

As is shown in Fig. 1, some guard cells are set to ensure that the reference cells do not contain targets. Some reference cells adjacent to the guard cells are selected to estimate the clutter power. The threshold is the product of a scale factor and the estimated power of clutter. If the power of signal in the test cell exceeds this threshold, we consider that a target is present. The CA-CFAR detector is easier to implement than many other complex detectors and it can detect the dominant scatterers directly with the low computation cost.

B. Feature Extraction

In a high-resolution radar system, the electromagnetism characteristic of targets can be seen as the scattering center model because the target size is much larger than the wavelength of radar signal [7]. We suppose that the transmitted signal is $s(t)e^{j\omega_c t}$, where $s(\bullet)$ is the signal envelope and ω_c is the carrier angular frequency. Then in the baseband, the u th returned echo of the e th scatterer from the f th range cell can be expressed as

$$z_{ufe}(t) = \sigma_{fe} s\left(t - \frac{2R_{ufe}}{c}\right) \exp\left(-j\frac{4\pi}{\lambda}R_{ufe}\right) \quad (1)$$

where λ represents the radar wavelength, R_{ufe} represents the radial distance between the radar and the e th scatterer in the f th range cell of the u th returned echo and σ_{fe} represents the strength of the e th scatterer in the f th range cell. From (1), we can see that the returned echo of each scatterer in a range cell is a complex value related to time. Then, the u th returned echo from the f th range cell $z_{uf}(t)$ is the coherent summations of the complex time returns from target scatterers in the f th range cell, namely $z_{uf}(t) = \sum_{f=1}^{V_f} z_{ufe}(t)$ with V_f denoting the number of target scatterers in the f th range cell.

Since only the specific f th range cell in u th echo is considered, the fast time t is a constant and can be omitted

$$z_{uf} = \sum_{f=1}^{V_f} \sigma_{fe} \exp\left(-j\frac{4\pi}{\lambda}R_{ufe}\right). \quad (2)$$

The HRRP $\mathbf{z}_u$ is expressed as $\mathbf{z}_u = [|z_{u1}|, |z_{u2}|, \dots, |z_{uF}|]^T$ with $|\bullet|$ and $[\]^T$ representing modulus and transpose operation. Note that the extracted feature samples are the intensities and positions of range cells containing dominant scatterers. Due to the huge intensity difference between dominant scatterer and weak scatterers, the intensity of a range cell containing dominant scatterer mainly depends on the intensity of the dominant

scatterer. Therefore, the extracted range cells containing dominant scatterers can be seen as the dominant scatterers approximately.

Fig. 2(a) to (i) shows five kinds of targets' and four kinds of artificial clutter's original HRRPs (left plots) and their dominant scatterers with normalized intensity (right plots). We can see some differences in dominant scatterers distributions of targets and clutter from Fig. 2. First, the dominant scatterers of some clutter with simple construction are less than those in targets. Second, the dominant scatterers distributions of targets and clutter with complex construction are different. For example, targets have more relatively strong dominant scatterers, whose intensities exceed half of the strongest scatterer's, than the clutter with complex construction.

In this article, the feature sample $\mathbf{S}$ is defined as

$$\mathbf{S} = \{\mathbf{s}_1, \mathbf{s}_2, \dots, \mathbf{s}_G\} \quad (3)$$

where $\mathbf{s}_g = (Int_g, Pos_g)$ is a two-dimensional point with Int_g and Pos_g representing the intensity and position of the g th dominant scatterer, $g = 1, 2, \dots, G$ and G denotes the number of dominant scatterers. Fig. 3 shows the flowchart of our feature extraction method. First, an HRRP fragment $\mathbf{z}$ is input to CA-CFAR processor. Undetected range cells are set to 0 and $\mathbf{z}_{det}$ is obtained. Second, $\tilde{\mathbf{z}}_{det}$ is obtained by normalizing the amplitude of $\mathbf{z}_{det}$. The purpose of normalization is to deal with the amplitude-scale sensitivity of HRRP data [13]. Third, $\tilde{\mathbf{z}}_{det}$ is shifted with its centroid locating at the mid-position and $\tilde{\mathbf{z}}_{det}^*$ is obtained. The purpose of alignment is to compensate the radial displacement in consecutive HRRPs. Finally, the $\mathbf{S}$ is obtained by extracting the intensities and positions of dominant scatterers in $\tilde{\mathbf{z}}_{det}^*$.

In this article, the alignment method is based on the dominant scatterers. Some comparisons of our alignment method with some traditional alignment methods are given in Section IV-C.

III. REJECTION METHOD BASED ON ONE-CLASS CLASSIFIER

A. Hausdorff Distance

The HD between two finite point sets $\mathbf{A} = \{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_D\}$ and $\mathbf{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_I\}$ is defined as

$$H(\mathbf{A}, \mathbf{B}) = \max(h_{LTS}(\mathbf{A}, \mathbf{B}), h_{LTS}(\mathbf{B}, \mathbf{A})) \quad (4)$$

where h_{LTS} is called the least trimmed square HD (LTS-HD) [14], and it is defined as

$$h_{LTS}(\mathbf{A}, \mathbf{B}) = \frac{1}{D} \sum_{\mathbf{a}_d \in \mathbf{A}} \min_{\mathbf{b}_i \in \mathbf{B}} M(\mathbf{a}_d, \mathbf{b}_i) \quad (5)$$

where $M(\mathbf{a}_d, \mathbf{b}_i)$ is the Mahalanobis distance between $\mathbf{a}_d$ and $\mathbf{b}_i$ with $d = 1, 2, \dots, D$ and $i = 1, 2, \dots, I$. The Mahalanobis distance $M(\mathbf{a}_d, \mathbf{b}_i)$ is defined as

$$M(\mathbf{a}_d, \mathbf{b}_i) = \sqrt{(\mathbf{a}_d - \mathbf{b}_i)^T \Sigma^{-1} (\mathbf{a}_d - \mathbf{b}_i)} \quad (6)$$

where Σ represents the covariance matrix of the feature vector. The Σ should be estimated by feature samples.

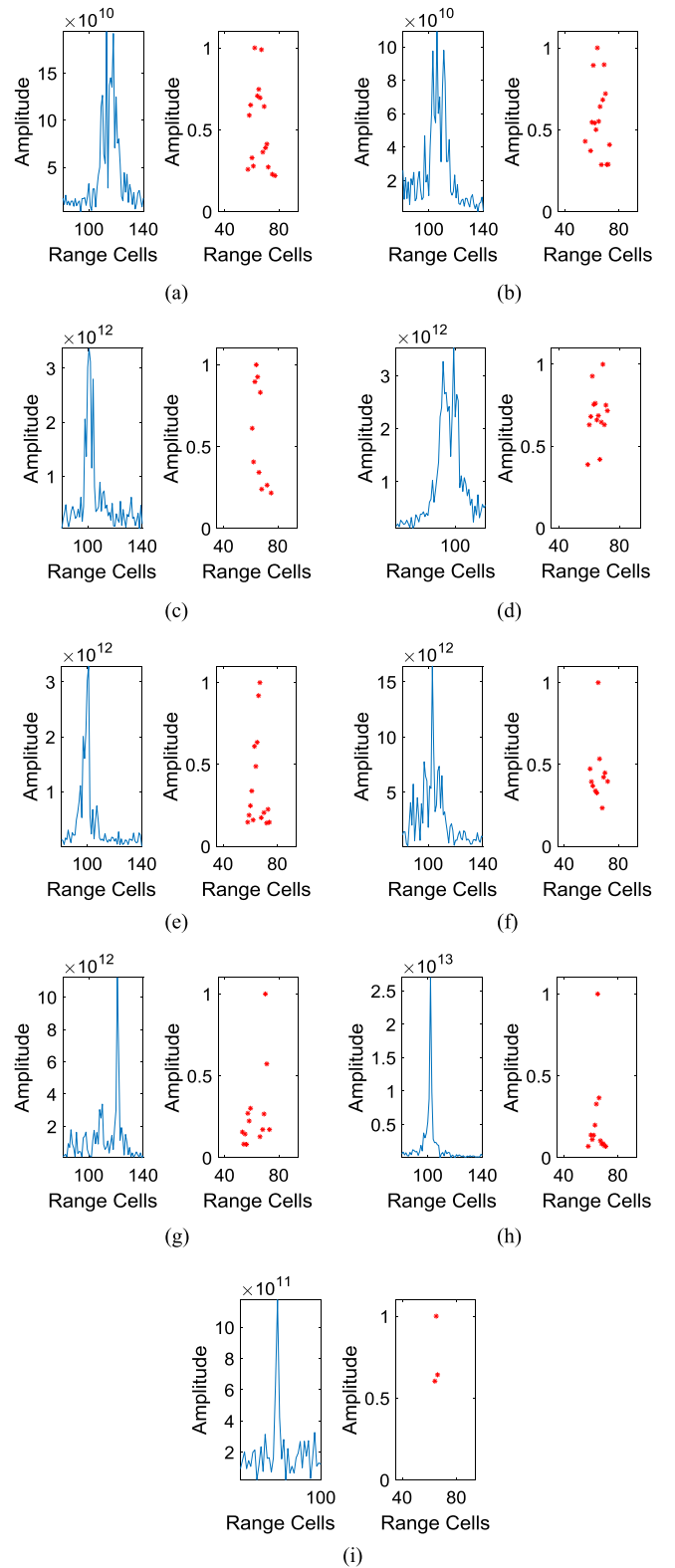


Fig. 2. Examples of some HRRP samples and their dominant scatterers [left plots in (a) to (i): original HRRPs; right plots in (a) to (i): dominant scatterers with normalized intensity]. (a) Tank A. (b) Tank B. (c) Model of tank C. (d) Troop carrier. (e) Truck. (f) Building. (g) Pylon. (h) Billboard. (i) Corner reflector.

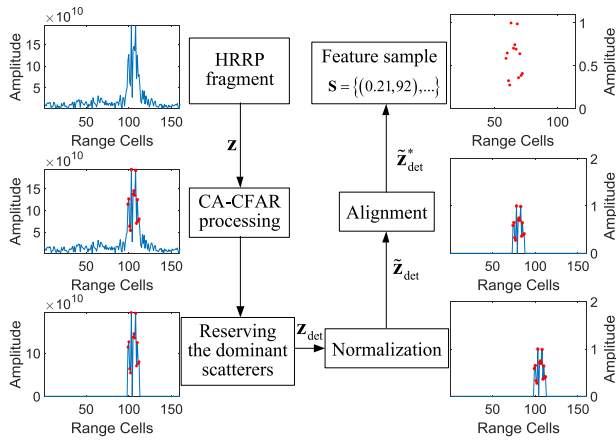


Fig. 3. Flowchart of the proposed feature extraction method.

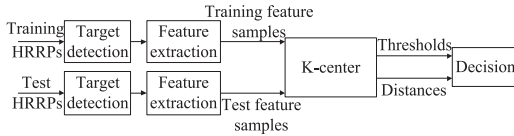


Fig. 4. Flowchart of proposed rejection method.

B. Rejection Method Based on One-Class Classifier Combined With HD

Fig. 4 shows the flowchart of the proposed rejection method. In training step, a threshold based on HD is obtained. The training samples are all from the target samples. In test step, whether the test sample is accepted is decided.

The process of K-center based on HD is introduced as follows. In training step, the training samples are divided into K clusters, of which K optimal center feature samples $\mathbf{o}_1, \mathbf{o}_2, \dots, \mathbf{o}_K$ and K corresponding thresholds are obtained. K optimal center feature samples can be obtained by the following procedures. First, K samples are selected as the initial center feature samples, $\mathbf{o}'_1, \mathbf{o}'_2, \dots, \mathbf{o}'_K$, of K clusters. Second, each training feature sample is assigned to its nearest cluster. A feature sample $\mathbf{x}$ is assigned to the m th cluster according to

$$m = \arg \min_{k=1,2,\dots,K} H(\mathbf{x}, \mathbf{o}'_k). \quad (7)$$

Third, new center samples in K cluster are found. Assuming w samples $\mathbf{x}_{m1}, \mathbf{x}_{m2}, \dots, \mathbf{x}_{mw}$ are in the m th cluster, the new center feature sample $\tilde{\mathbf{o}}'_m$ of the m th cluster is

$$\tilde{\mathbf{o}}'_m = \mathbf{x}_{mn'}, \quad n' = \arg \min_{n=1,2,\dots,w} \left[\max_{j=1,2,\dots,w} H(\mathbf{x}_{mn}, \mathbf{x}_{mj}) \right]. \quad (8)$$

Finally, the second and the third procedures are repeated until the K center feature samples do not change.

Then, the threshold Thr_m of the m th receptive cluster containing w samples can be obtained as follows. First, the HDs between the center sample to other $w - 1$ samples in the m th cluster are sorted in ascending order to compose $\mathbf{H}_m$: $\mathbf{H}_m = [H_{m1} \cdots H_{mw-1}]$. Second, the value of index n is

set according to an adjustable parameter p and the number of feature samples w : $n = \text{round}(w \cdot (1 - p))$, where $\text{round}(\bullet)$ represents the rounded-off value operator. Finally, the Thr_m can be obtained by selecting the n th element in $\mathbf{H}_m$: $Thr_m = H_{mn}$. Here, p is the rejection rate of training samples with its values in the range of 0 to 1. The larger the p is, the more the training samples are discarded. In practice, p and K need to be set *a priori*. Note that different clusters in K-center do not necessarily account for different targets. On the one hand, due to the presence of target-aspect sensitivity, the HRRPs from different aspects of a target may be divided into different clusters. On the other hand, the samples from different targets may be similar in some specific aspects. These similar samples may gather in the same cluster.

In the test step, the HD between the test feature sample $\mathbf{y}$ and its nearest cluster can be calculated as

$$H_{test}(\mathbf{y}) = \min_{m=1,2,\dots,K} H(\mathbf{y}, \mathbf{o}_m), \quad m_{test} = \arg \min_{m=1,2,\dots,K} H(\mathbf{y}, \mathbf{o}_m) \quad (9)$$

where m_{test} denotes the index of the nearest cluster of $\mathbf{y}$. If $H_{test}(\mathbf{y}) \leq Thr_{m_{test}}$, the test sample will be accepted.

IV. REJECTION EXPERIMENTS ON MEASURED DATA

A. Introduction of the Measured Data

The measured data are collected by a high-resolution radar with the range resolution being 0.293 m. 11420 HRRPs are obtained totally, of which 2631 HRRPs are from tank A, 4586 HRRPs are from tank B, 64 HRRPs are from the model of tank C, 84 HRRPs are from the troop carrier, 731 HRRPs are from the truck, 400 HRRPs are from the building, 731 HRRPs are from the pylon, 1462 HRRPs are from the billboard, and 731 HRRPs are from the corner reflector. The average signal-to-noise ratios (SNRs) of the original data of tanks A, B, model of C, troop carrier, truck, building, pylon, billboard, and corner reflector are about 23.31, 19.66, 18.36, 21.89, 24.62, 24.08, 24.13, 19.63, and 19.47 dB, respectively. In this article, the Tank A, Tank B, the model of Tank C, the troop carrier and the truck are targets of interest and others are outliers.

Fig. 5 shows the measured data and some processed results. In Figs. 5(a) to (e), the first plot show the original HRRPs and the second plot show the extracted dominant scatterers with normalized intensity. For tanks A and B, the radar works in the tracking mode. Fig. 6 shows the movements of tanks A and B. For tank A, at first it moves perpendicular to the radar sight from 200 m away. Then, it moves away from radar at an almost constant speed. And for tank B, at first it moves toward the radar from about 220 m away. Then it turns around and then moves away. The model of tank C, the troop carrier and the truck stay still with their front sides toward radar. For these three targets, the radar works in scan mode with a constant pitch angle and the azimuth angle in the range of -10 to 10° . Fig. 5(f) to (i) shows the samples after normalization and alignment from different kinds of artificial clutter. For the four kinds

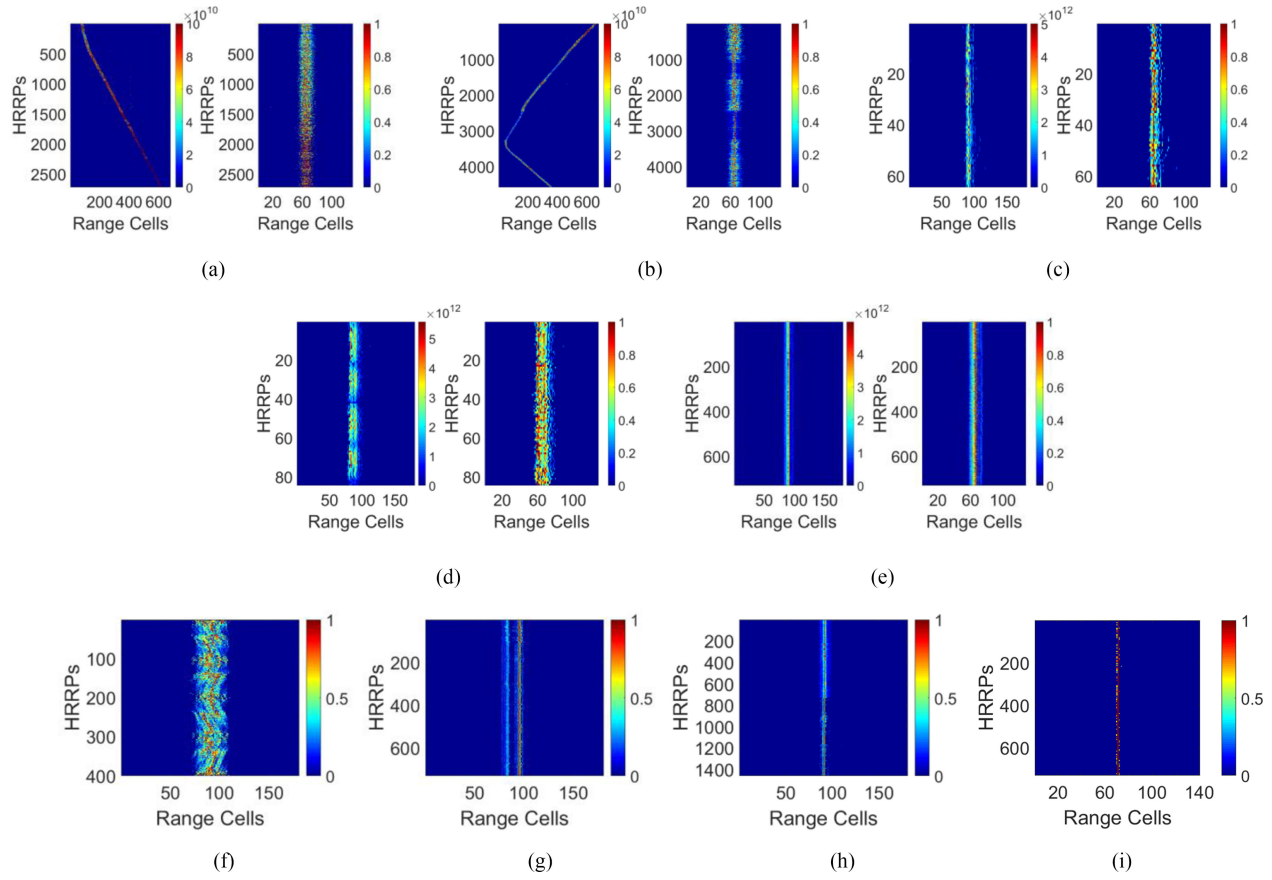


Fig. 5. Data of the five kinds of targets and artificial clutter [the left plots in (a) to (e) are the original HRRPs, the right plots in (a) to (e) and the plots in (f) to (h) are the HRRPs after alignment and normalization]. (a) Tank A. (b) Tank B. (c) Model of tank C. (d) Troop carrier. (e) Truck. (f) Building. (g) Pylon. (h) Billboard. (i) Corner reflector.

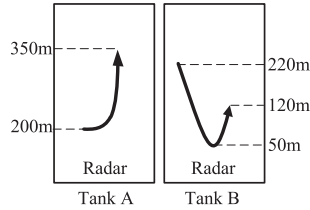


Fig. 6. Diagrams of the movements of tanks A and B.

of artificial clutter, the radar works with constant azimuth and pitch angles.

The parameters of CA-CFAR are set as follows. Here, the lengths of targets are about 7 to 10 m, corresponding to 23 to 34 range cells. Therefore, we set 35 guard cells in the leading and lagging, respectively. According to the measurement scene, the number of reference cells is set to 60. Through some experiments with the measured data samples, we find that the feature extraction results change little with false alarm rate. Then, the false alarm rate is set to a moderate value 10^{-4} .

B. Experiments Based on different Rejection Methods

Here, we adopt the amplitude of the HRRP, the amplitude of frequency spectrum (AFS) and the power spectrum

(PS) in [21], combined with the K-center, K nearest neighbor (KNN) [9] and linear support vector domain description (L-SVDD) [26] as the comparison methods. In addition, KNN combined with our proposed feature is also adopted as a comparison method. The amplitude, the AFS and the PS of an HRRP are written as

$$\begin{aligned} \mathbf{x}_{amp} = \mathbf{z} &= [|z_1|, |z_2|, \dots, |z_F|], \quad \mathbf{x}_{AFS} = |\text{DFT}(\mathbf{z})|, \\ \mathbf{x}_{PS} &= |\text{DFT}(\mathbf{z})|^2 \end{aligned} \quad (10)$$

where $\text{DFT}(\bullet)$ represents the discrete Fourier transform. The PS, AFS, and amplitude features from tank A are given in Fig. 7. The left column in Fig. 7 shows the extracted feature samples from all the HRRPs of tank A and the right column shows the feature samples from 10th HRRP of tank A.

The process of KNN one-class classifier is introduced as follows. In training step, every training sample in the feature space corresponds to a hypersphere area with itself as the center and the distance between it to its l th nearest samples as the radius. In test step, a test sample is accepted when it belongs to any hypersphere in feature space. The l is an adjustable parameter and needs to be set *a priori* in practice.

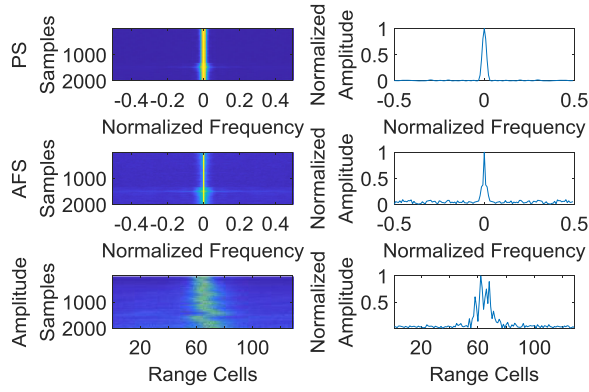


Fig. 7. PS, AFS, and amplitude features of HRRPs from Tank A (the left column: extracted feature samples from tank A; right column: the extracted feature samples from tenth HRRP of tank A)

The process of L-SVDD is introduced as follows. The training step can be seen as an optimal problem

$$\begin{aligned} \min_{rad, cen, \zeta_i} \quad & rad^2 + v \sum_i \zeta_i \\ \text{s.t.} \quad & \|\mathbf{x}_i - cen\|^2 \leq rad^2 + \zeta_i, \zeta_i \geq 0, \forall i \end{aligned} \quad (11)$$

where $\|\cdot\|$ is the l_2 norm, cen is the center, rad is the radius, $\mathbf{x}_i$ is the i th training sample, ζ_i is the slack variable and v is the penalty factor. By solving this optimal problem, the center and the radius of a hypersphere in feature space can be obtained. In test step, a test sample will be accepted when it belongs to the hypersphere in feature. Here, the v is an adjustable parameter with its range of 0 to 1. Note that our proposed feature combined with HD cannot be used in L-SVDD. That is because the optimal problem in L-SVDD cannot be solved when the proposed feature with HD is introduced.

In training step, 1000 samples from tank A, 1000 samples from tank B, 32 samples from the model of tank C, 42 samples from troop carrier and 365 samples from truck are selected. In test step, other 1000 samples from tank A, 1000 samples from tank B, 32 samples from the model of tank C, 42 samples from troop carrier and 365 samples from truck and all the clutter samples are selected. The training and test target samples are selected randomly. Each rejection process is repeated 100 times. In this article, the average accurate rate is defined to evaluate the rejection performance

$$Pr_{ave} = (Pr_{f+} + Pr_{t+}) / 2 \quad (12)$$

where Pr_{f+} is the clutter accurate rate, Pr_{t+} is the targets accurate rate.

1) *Rejection Results of Different Rejection Methods:* The parameters K , p in K-center, the l in KNN, and the v in L-SVDD need to be chosen *a priori*. The ten-fold cross validation is adopted to select the parameters. Here, 25% clutter samples are selected randomly for cross validation process. The target samples are divided into ten equal parts randomly. For each time nine parts are used for training and the remaining one part and the selected clutter samples are used for validation under the different

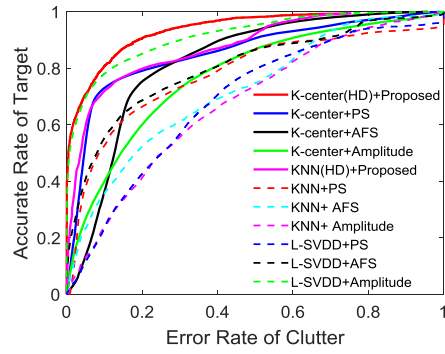


Fig. 8. ROC curves of different rejection methods.

values of parameters. The process is repeated ten times. The parameter value with best average performance during the ten validation processes is selected. Here, we set the parameters $p \in [0, 0.99]$ with intervals of 0.01, $K \in [1, 10]$ with intervals of 1, $l \in [1, 2439]$ with intervals of 1 and $v \in [0, 0.99]$ with intervals of 0.01.

Table I gives the selected values of parameters using ten-fold cross validation and the rejection results over 100 experiments under these values of parameters. As given in Table I, the proposed feature combined with HD performs better than other features for K-center and KNN, which indicates that the proposed feature provide more abundant information. From Table I, we can also see that our proposed method is better than the KNN combined with proposed feature and HD, which means the K-center can describe the distribution of training feature samples better than KNN. Besides, the proposed method performs the best among all the rejection methods.

The receiver operator characteristic (ROC) curve and the area under curve (AUC) are good measures for rejection performance [22]. Fig. 8 and Table II give the mean ROC curves and the mean AUCs of different methods over 100 experiments.

From Fig. 8, we can see that the ROC curve of our proposed method is above other ROC curves when the outlier accepted rate is lower than 73.13%. Since the purpose of rejection is to eliminate the false alarms, the outliers accepted rate will not be set high in practice. Therefore, the proposed method is suitable for application. From Table II, we can see that the proposed method has the highest AUC, which also indicates that the proposed method performs the best. Besides, we can see from Fig. 8 and Table II that the proposed feature is the best among all the features for K-center and KNN.

2) *Rejection Results Under different SNRs:* Then, we will discuss the rejection results under different SNRs. The SNR is defined as [5]

$$SNR = 10 \log_{10} \left(\frac{\sum_{i=1}^N P_t(i)}{N \times \bar{P}_n} \right) \quad (13)$$

where $P_t(i)$ is the target (or artificial clutter) power in i th target (or artificial clutter) range cell, $\bar{P}_n$ is the noise average power, N is the number of the range cells occupied by

TABLE I
Rejection Results of Different Rejection Methods Under the Selected Values of Parameters (Mean Values and Standard Deviations Are Obtained over 100 Experiments)

Rejection method	Accurate rate of targets (%)	Accurate rate of clutter (%)	Average accurate rate (%)	Parameters (10-fold cross validation)
K-center(HD)+Proposed feature	93.921±0.010	77.243±0.063	85.582±0.015	$p = 0.09, K = 3$
K-center+PS	76.024±0.043	86.243±0.493	81.134±0.165	$p = 0.30, K = 4$
K-center+ AFS	80.379±0.043	74.065±0.129	77.222±0.039	$p = 0.23, K = 6$
K-center+Amplitude	75.623±0.114	67.157±0.619	71.390±0.169	$p = 0.41, K = 6$
KNN(HD)+proposed feature	75.279±0.014	87.181±0.020	81.230±0.004	$l = 2089$
KNN+PS	58.961±0.003	87.277±0.003	73.574±0.013	$l = 1662$
KNN+ AFS	42.207±0.019	86.131±0.013	64.169±0.006	$l = 594$
KNN+Amplitude	35.686±0.015	87.394±0.009	61.515±0.004	$l = 279$
L-SVDD+PS	67.338±0.002	57.734±0.024	62.536±0.007	$v = 0.32$
L-SVDD+ AFS	67.402±0.014	81.592±0.005	74.497±0.010	$v = 0.43$
L-SVDD+Amplitude	78.831±0.015	88.228±0.002	83.529±0.002	$v = 0.20$

TABLE II
AUCs of Different Rejection Methods

Rejection method	AUC
K-center(HD)+Proposed feature	0.9369
K-center+PS	0.8563
K-center+ AFS	0.8227
K-center+Amplitude	0.7736
KNN(HD)+Proposed feature	0.8767
KNN+PS	0.7779
KNN+ AFS	0.8227
KNN+Amplitude	0.6763
L-SVDD+PS	0.6868
L-SVDD+AFS	0.8018
L-SVDD+ Amplitude	0.9161

the target (or artificial clutter). Here, N is set according to the number of extracted scatterers. Since the SNRs of original data of different targets or clutter are different, the denoising method based on orthogonal matching pursuit algorithm [27] is implemented for original HRRPs. The SNRs of denoised data are larger than 40 dB. Then five noisy data sets are generated by adding Gaussian white noise to the denoised HRRPs. The SNRs of five noisy data sets are 25, 20, 15, 10, and 5 dB, respectively. The selection of training and test samples is similar to that in Section IV-B. The training samples are from the denoised data and the test samples are from the noisy data. Fig. 9 gives the mean performances of different method with SNR over 100 experiments. As shown in Fig. 9, the proposed rejection method is better than others when $\text{SNR} \geq 10$ dB. When $\text{SNR} = 5$ dB, all rejection methods perform badly because

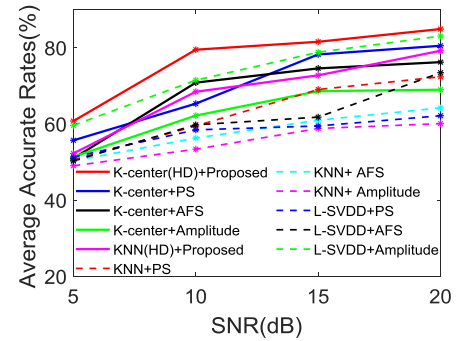


Fig. 9. Average accurate rates with SNR.

of the relatively strong noise. Besides, we can also see from Fig. 9 that the proposed feature is the best against noise among all the features for K-center and KNN.

3) *Rejection Results Under different Parameter Values:* Here, we will discuss the change of rejection performance with parameter values. We select two best methods based on KNN and L-SVDD, the KNN combined with the proposed feature and L-SVDD combined with amplitude, to compare with the proposed method. The parameter l has different scale from p and v . Therefore, we use the $l/2439$ to replace l when comparing their performances.

Fig. 10 shows the mean rejection performances of three rejection methods under different parameter values over 100 experiments. Fig. 10(a) shows the average accurate rates of our method when $p \in [0.03, 0.52]$ and $K = 3$, Fig. 10(b) shows the average accurate rates of KNN combined with proposed feature when $(l/2439) \in [0.48, 0.99]$, and Fig. 10(c) shows the average accurate rates of L-SVDD combined with amplitude when $v \in [0.01, 0.50]$. In Fig. 10(a)–(c), the means of average accurate rates are 0.8076, 0.7482, and 0.7863, and the variances of average

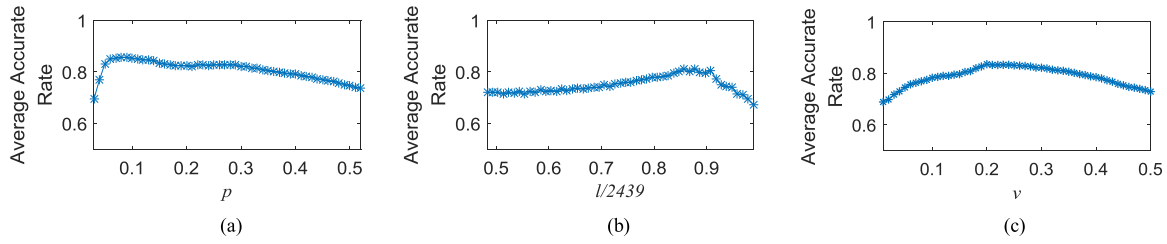


Fig. 10. Average accurate rates of different rejection methods under different parameter values. (a) The proposed method versus p . (b) KNN combined with proposed feature versus l . (c) L-SVDD combined with amplitude versus v .

TABLE III
Mean difference Parameter Values of Different Alignment Methods Under Different SNRs

Alignment methods	10dB	15dB	20dB
Proposed	1.17	0.54	0.31
Centroids of HRRPs	32.81	16.37	8.72
Envelope cross-correlation	5.13	1.56	0.35

accurate rates are 0.0013, 0.0011, and 0.0014. From Fig. 10, we can see that the rejection performances degrade when the parameters are not set to the optimal value. The variances of average accurate rates of three methods are almost the same. Our proposed method performs the best nearby the optimal parameter value.

C. Effectiveness of Proposed Alignment Algorithm

Here, we will compare the alignment method with the other two traditional alignment methods. One is the alignment based on the centroids of HRRPs, and another one is the envelope cross-correlation method [8]. The denoised data and three noisy data sets of tank A are adopted. The SNRs of three noisy data sets are 20, 15, and 10 dB. First, we define a shift value vector $\mathbf{sh} = \{sh_1, sh_2, \dots, sh_C\}$ to describe the shift value of every HRRP in alignment process, where C represents the number of HRRPs to be alignment. Then we let $\mathbf{sh}'$ represents the shift value vector using denoised data and $\mathbf{sh}''$ represents the shift value vector using noisy data. Due to the high SNR, the alignment results using denoised data can be seen approximately ideal. Here, an index, the mean difference parameter dif , is defined to describe the alignment performance difference in using denoised data and noisy data

$$dif = \frac{1}{C} \|\mathbf{sh}' - \mathbf{sh}''\|_1 \quad (14)$$

where $\|\cdot\|_1$ represents the one-norm. The smaller the dif is, the smaller the performance difference between noisy and ideal situations is, and thus the better the alignment performance is. Table III gives the mean difference parameter values of different alignment methods under different SNRs. From Table III, we can see that performance of the proposed method degrades the least among three alignment methods. That is because the dominant scatterers have relatively strong intensities and thus the extraction of dominant scatterers suppresses the noise to some extent.

V. CONCLUSION

In this article, a rejection method is introduced to eliminate the false alarms that emerge during the detection stage. In our method, the intensities and positions of dominant scatterers are extracted. The K-center with HD is adopted as the one-class classifier. Experimental results on measured data indicate that our rejection method performs better than some other methods. Meanwhile, our proposed rejection performs well against noise.

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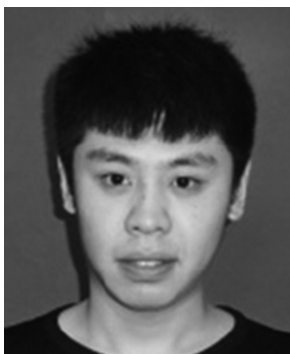
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